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Task Assignment in Crowdsourcing Using Deep Knowledge Tracing

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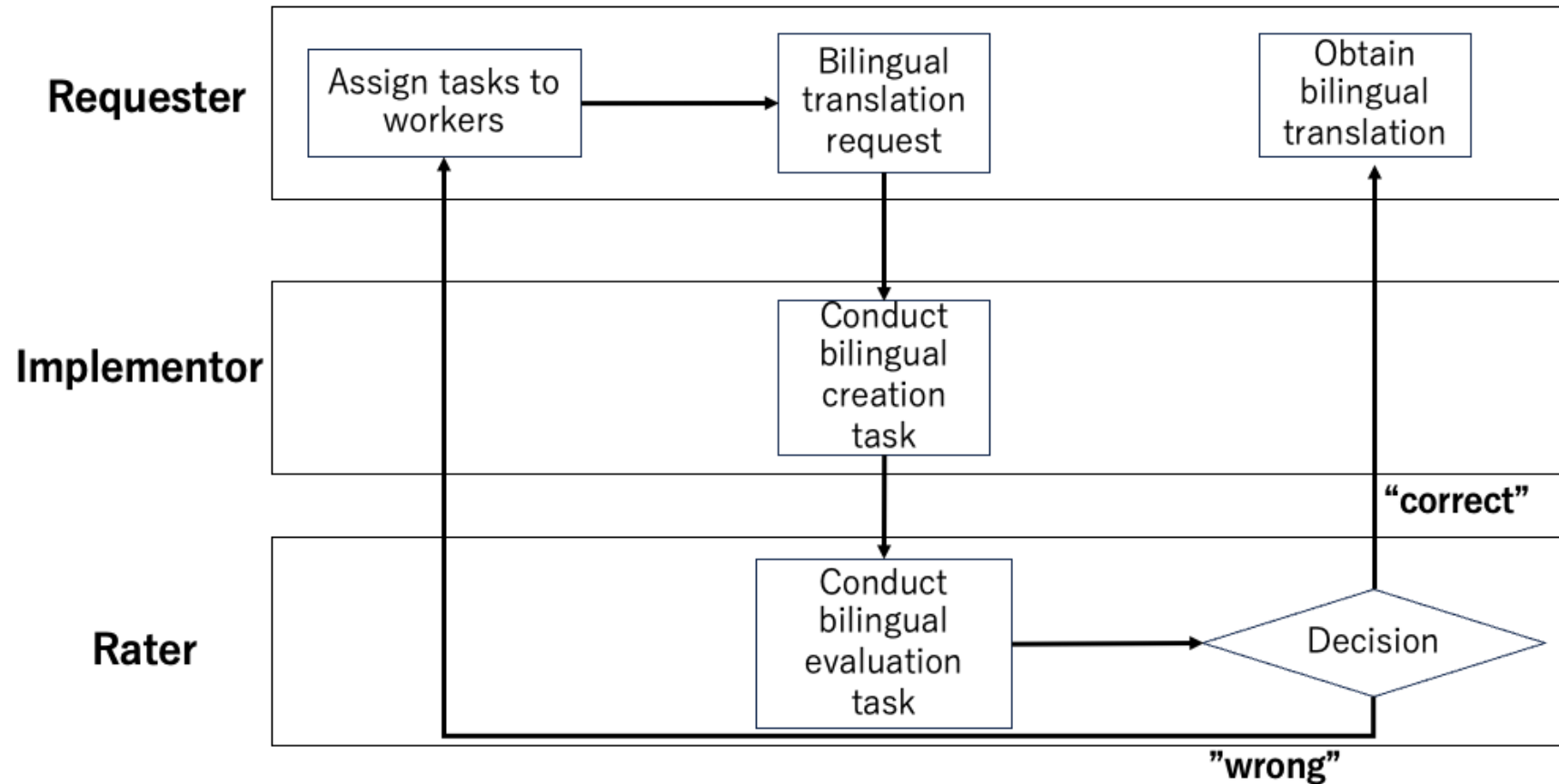
Background



- There are over 700 ethnic languages in Indonesia
 - 147 languages are threatened with extinction
 - Bilingual dictionaries are needed to support communication
- ↓
- Bilingual dictionaries are being manually created through crowdsourcing

Problems

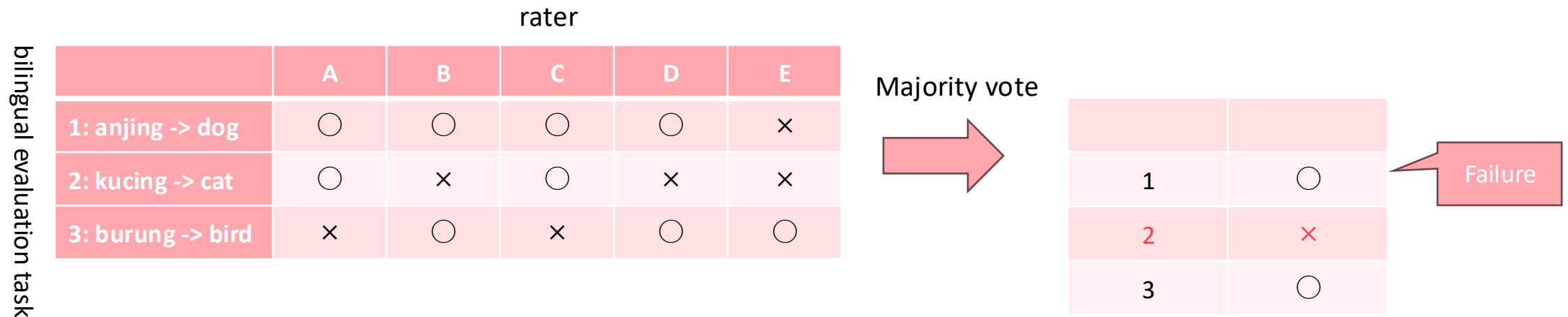
- Cost increase due to mismatch between work and task



Problems

Challenges in low-resource languages

- Lack of skilled workers
- Ensuring translation quality is difficult



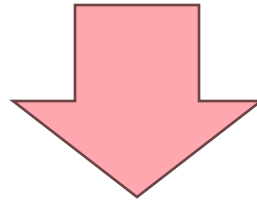
Purpose

Goals

- Enhance individual worker accuracy
- Match tasks to worker's skills
- Optimize assignments to reduce rework and time

Proposed Method

- Predict task success from past work history
 - Predict task success rate with **Deep Knowledge Tracing (DKT)**



- Used for task assignment considering the difficulty of the task
 - **Extended Hungarian method** for many-to-many assignment

Deep Knowledge Tracing(DKT)

- Knowledge Tracing(KT)
 - Predict learner's knowledge level
 - Mainly used in the field of education
 - Success probabilities (0~1)
- LSTM is used for KT
 - No need to manually enter domain knowledge
 - Allows long-term dependency learning

Deep Knowledge Tracing(DKT)

Input data

- Worker ID, knowledge tag, task result (success or failure)

Model operating principle

- Predict next task success probability from work history at **time t**
- Use LSTM to sequentially accumulate past history

Update hidden layer

$$h_t = f(W_{ih}x_t + W_{hh}h_{t-1} + b_h)$$

x_t : Input at time t

h_t : State of hidden layer

f : Activation function

Output layer calculation

$$y_t = g(W_{hp}h_t + b_o)$$

y_t : Success probability of next task ($0 \leq y_t \leq 1$)

g : Sigmoid function

Category

Role of category

- Classify tasks by skill and knowledge
- Same tags imply similar skills

Necessiity in this study

- Required for DKT prediction
- Tags must be created as none exist in the data

Category: How to categorize

- Data from the Indonesian and Minangkabau translation evaluation tasks
 - Indonesian words: 986 words
- 1. Use Word2Vec to convert the task words into 300 dimensional vectors
- 2. Clustering the vectors using the k-means method
- 3. Using the clustering results as category
 - Make sure that the number of elements in each cluster is 5 or more
 - We calasfied indonensian words into 10 categories

Category

Class	0	1	2	3	4
Number of elements	57	105	124	53	102

Class	5	6	7	8	9
Number of Elements	10	108	60	263	104

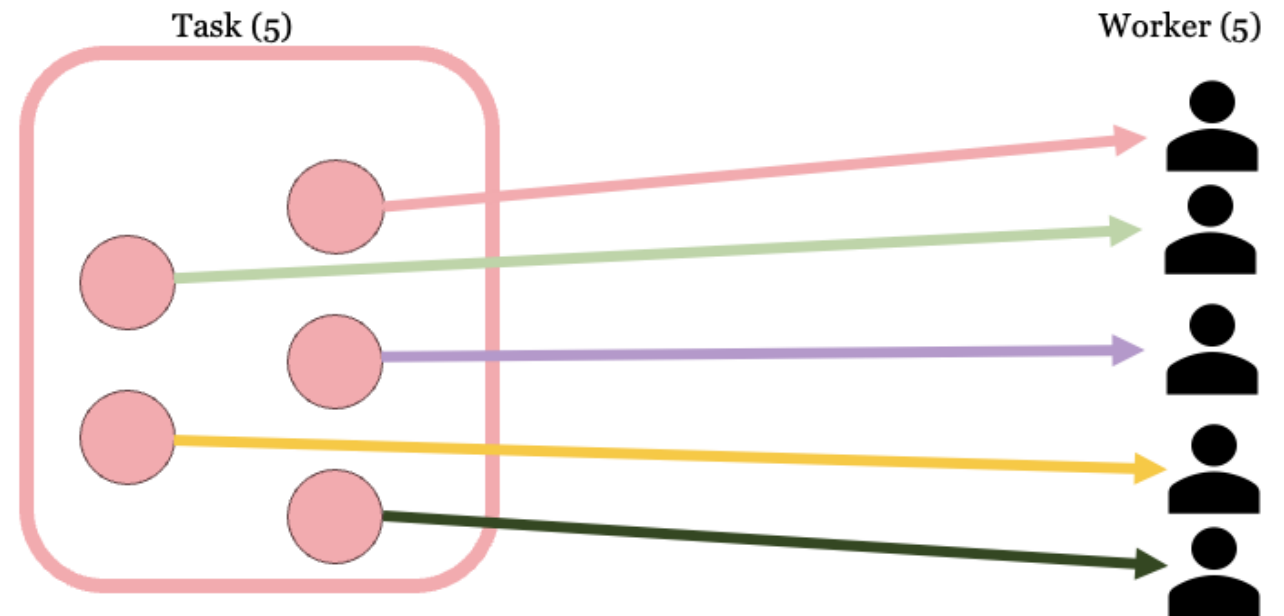
- 10 types (0 ~ 9) of category were assigned to 986 words
- Each category could be classified as roughly regular
 - Time and order (class 0), verbs (class 1), relationships (class 8), etc.

Task Allocation

Hungarian method

- Allocation that minimizes cost between limited workers and tasks
- Input: Cost matrix
- Output: One optimal task assigned to each worker
- This is limited to one-to-one assignment problems

		Cost matrix Worker ID				
Task ID		1	2	3	4	5
	1	9	2	7	8	6
	2	6	4	3	7	8
	3	5	8	1	8	7
	4	7	6	9	4	2
	5	8	3	6	2	9



Task Assignment: Extended Hungarian Method

Crowdsourcing issues

- Many-to-many assignment required
- Number of workers $\neq$ number of tasks

Extended approach

- Square matrix conversion of the cost matrix

Inversion of success probability

- Treat success probability as cost and find the combination maximizing total cost

Experiment: Outline of experiment

Purpose

- Verification of the effect of the proposed method (allocation method using DKT)
- Comparison with other allocation methods and evaluation of accuracy

Comparison method

- Assignment method using DKT (proposed method)
- Ratio-based assignment
- Random allocation method

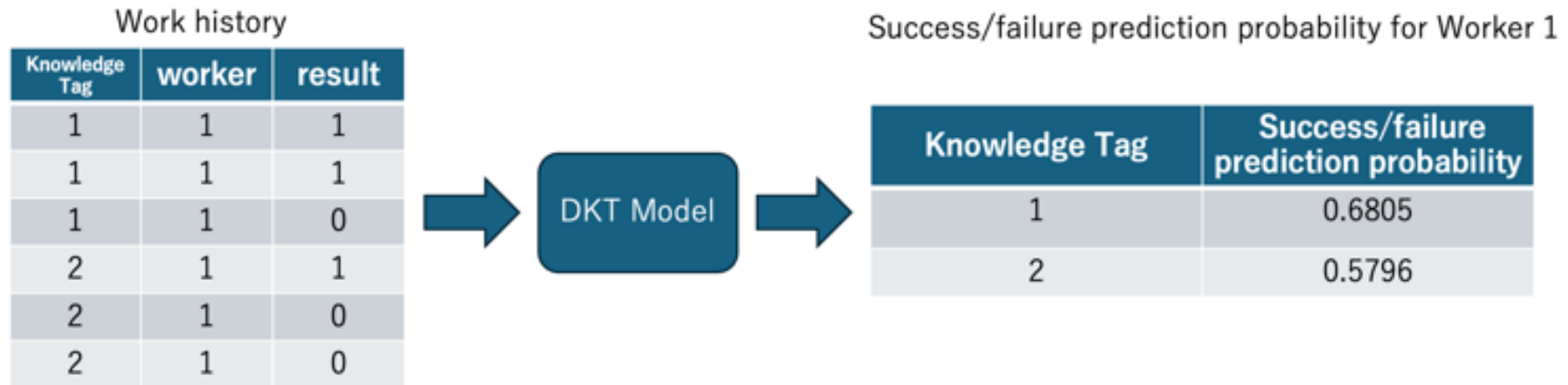
Experiment: Data used in the experiment

Data for Experiment

- Training data: 1450 tasks per worker
- Category: 10 categories
- Worker: 20 workers
- Test data: 500 tasks
- Ensures balanced category distribution

Experiment: Assignment Method Using DKT

- Trained Model with 1450 Work Histories
- Assigned 500 tasks to 20 workers using DKT model predictions
(25 tasks per worker)



Experiment: Method using Actual Success Ratio and Random assignment method

Ratio-based assignment

- Calculates success ratio of past 1450 tasks for each category
- Assigning 500 tasks to 20 workers using the ratio

Random assignment

- Randomly assign 500 tasks 5 times
- Average the percentage of correct answers 5 times for each worker

Results

- 2.4% better than Ratio-based assignment
- 5.6% better than the random assignment

Workers 1, 2, 6, 9, 14, 17, 18, 20

- The proposed method has the highest accuracy

Worker 16

- The proposed method has the lowest accuracy

Average of allocation results

	Highest percentage of correct answers	Lowest percentage of correct answers	Average
DKT	96.0%	64%	75.4%
Ratio-based	92%	56%	73.0%
Random			69.8%

Discussion: Confidence level in the worker's work

Confidence level

- Indicates the reliability of worker’s result
- The higher the number, the better the worker
- The accuracy was generally improved at all confidence levels

Confidence level		DKT(%)	Ration-based(%)	Random (%)
	1	71.0	71.0	64.4
	2	74.0	76.0	63.6
	3	76.8	76.8	71.0
	4	72.0	60.0	65.6
	5	82.0	77.0	83.0

Discussion: Issues

Problem of the number of data used in the experiment

- Current data for DKT: 1450 , tasks: 500
- More data improves DKT accuracy

How to classify category

- Word2vec used for classification
- Cluster size imbalance affects accuracy
- Small Clusters are hard to predict

Summary

Research results

- The proposed method improves task assignment accuracy in crowdsourcing
- Improves efficiency of bilingual dictionary creation for low-resource languages

Future prospects

- Expansion of data used in experiments and comparison with other machine learning
- Application and feedback at actual crowdsourcing sites
- Get a prediction for each task

Thank You!